AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 120 Minutes

Instructions:

- Total Marks: 30
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

1. When a prediction model is trained using data that already includes the correct class labels, this type of learning is called _____?

A) Unsupervised Learning
B) Supervised Learning
C) Semi-Supervised Learning
D) Hybrid Supervised learning

2. If the class label is numeric values in a prediction task, it is called _____?

A) Classification
B) semi-supervised
C) hybrid supervised
D) Regression

3. The commonly used activation function for CNN is _____?

A) Linear
B) Sigmoid
C) ReLU (Rectified Linear Unit)
D) Tanh

4. They main objective/ purpose of Image classification is _____?

A) Detecting objects in an image
B) Identifying and assigning a label to an image
C) Reducing image resolution
D) Generating artistic images

5. SVM is a powerful and versatile technique that can be used for _____?

A) Linear Classification
B) Non-linear classification
C) Regression
D) All of these

Part – B (15 Marks)

Provide brief answers to the following questions.

1) Give at least three real-world examples where we can apply Deep Learning. (3 Marks)

2) Explain linear and non-linear regression (6 Marks)

3) What is SVM. Name at least three types of SVM. (6 Marks)

Part – C (10 Marks)

NOTE: This question does not need any programming code.

Imagine you are assigned to create an online learning recommendation app that suggests suitable courses to students based on their interests, academic background, skill level, and previous learning history. Provide a step-by-step solution for developing the recommendation system for this app.

Instructions:

- Provide a solution for building a recommendation system.
- Present your solution in a stepwise manner.
- Ensure that your response consists of a minimum of Five steps.

End of questions

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